

SECTION A: QUESTIONS

◆ Stack & Queue

1. Push sequence: 1, 2, 3, 4, 5
Pop sequence: 4, 5, 3, 2, 1
☞ Is this sequence valid?
 2. Operations: Push(5), Push(2), Push(8), Pop(), Push(1)
☞ Find the minimum element.
 3. Circular Queue: Size = 5, Front = 2, Rear = 1
☞ Is the queue full?
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♣ Trees

4. Inorder: D B E A F C
Preorder: A B D E C F
☞ Find the root node.
 5. Maximum nodes in a tree = 127
☞ Find the height.
 6. Insert into BST: 50, 30, 70, 20, 40, 60, 80
☞ Find inorder traversal.
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▮ Graphs

7. Directed graph with 4 nodes
☞ Find maximum number of edges.
 8. Graph: $A \rightarrow B, C$; $B \rightarrow D$; $C \rightarrow E$
☞ DFS traversal from A.
 9. Edges: $(A \rightarrow B)$, $(B \rightarrow C)$, $(C \rightarrow A)$
☞ Does a cycle exist?
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🔑 Hashing

10. $h(k) = k \bmod 7$
Insert: 10, 17, 24 using linear probing
☞ Find final positions.
11. $h_1(k) = k \bmod 10$, $h_2(k) = 7 - (k \bmod 7)$
Insert key 23
☞ Find position.
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▲ Heaps

12. Max heap: [50, 30, 40, 10, 20]
Delete root
☞ Find new root.
13. Insert 15 into min heap [10, 20, 30]
☞ Final heap?
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⚡ Complexity

14.
for $i = 1$ to n
 for $j = 1$ to i
☞ Find complexity.
15.
for $i = 1$ to n
 $j = n$
 while $j > 1$: $j = j/2$
☞ Find complexity.
16. Recurrence: $T(n) = 3T(n-1)$
☞ Find complexity.
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□ Mixed

17. Two stacks in one array of size 10
☞ Maximum elements stored?

18. Queue implemented using two stacks

☞ Worst-case dequeue complexity?

19. Expression: $\{([])\}$

☞ Valid or not?

20. Graph with $n = 6$, edges = 15

☞ Identify graph type.